

the null hypothesis is rejected if and only if $\Delta(\mathbb{X}_m, \mathbb{Y}_n) = 1$. The test is usually designed to control the probability of *type I error*

$$\sup_p \mathbb{P}_{p \times p}(\Delta(\mathbb{X}_m, \mathbb{Y}_n) = 1) \leq \alpha$$

for a given $\alpha \in (0, 1)$, where the supremum is taken over all probability densities on $\mathbb{R}^d$. We then say that the test has *level* α . For all the definitions, if the test Δ depends on other random variables, we take the probability with respect to those too. For a given fixed level α , the aim is then to construct a test with the smallest possible probability of *type II error*

$$\mathbb{P}_{p \times q}(\Delta(\mathbb{X}_m, \mathbb{Y}_n) = 0)$$

for specific choices of alternatives for which $p \neq q$. If this probability is bounded by some $\beta \in (0, 1)$, we say that the test has *power* $1 - \beta$ against that particular alternative. In the asymptotic framework, for a consistent test and a fixed alternative with $\|p - q\|_2 > 0$, we can find large enough sample sizes m and n so that the test has power close to 1 against this alternative. In the non-asymptotic framework of this paper, the sample sizes m and n are fixed. We can then find an alternative with $\|p - q\|_2$ small enough so that the test has power close to 0 against this alternative. Given a test Δ , a class of functions $\mathcal{C}$ and some $\beta \in (0, 1)$, one can ask what the smallest value $\tilde{\rho} > 0$ is such that the test Δ has power at least $1 - \beta$ against all alternative hypotheses satisfying $p - q \in \mathcal{C}$ and $\|p - q\|_2 > \tilde{\rho}$. Clearly, this depends on the sample sizes: as m and n increase, the value of $\tilde{\rho}$ decreases. This motivates the definition of *uniform separation rate* (Baraud, 2002)

$$\rho(\Delta, \mathcal{C}, \beta, M) := \inf \left\{ \tilde{\rho} > 0 : \sup_{(p, q) \in \mathcal{F}_{\tilde{\rho}}^M(\mathcal{C})} \mathbb{P}_{p \times q}(\Delta(\mathbb{X}_m, \mathbb{Y}_n) = 0) \leq \beta \right\}$$

where $\mathcal{F}_{\tilde{\rho}}^M(\mathcal{C}) := \{(p, q) : \max(\|p\|_\infty, \|q\|_\infty) \leq M, p - q \in \mathcal{C}, \|p - q\|_2 > \tilde{\rho}\}$. For uniform separation rates, we are mainly interested in the dependence on $m + n$: for example we will show upper bounds of the form $a(m + n)^{-b}$ for positive constants a and b independent of m and n . The greatest lower bound on the uniform separation rates of all tests with non-asymptotic level $\alpha \in (0, 1)$ is called the *minimax rate of testing* (Baraud, 2002)

$$\underline{\rho}(\mathcal{C}, \alpha, \beta, M) := \inf_{\Delta_\alpha} \rho(\Delta_\alpha, \mathcal{C}, \beta, M),$$

where the infimum is taken over all tests Δ_α of non-asymptotic level α for testing $\mathcal{H}_0 : p = q$ against $\mathcal{H}_a : p \neq q$, and where we compare uniform separation rates in terms of growth rates as functions of $m + n$. This is a generalisation of the concept of critical radius introduced by Ingster (1993a,b) to the non-asymptotic framework. A test is *optimal in the minimax sense* (Baraud, 2002) if its uniform separation rate is upper-bounded up to a constant by the minimax rate of testing. As the class of functions $\mathcal{C}$, we consider the Sobolev ball

$$\mathcal{S}_d^s(R) := \left\{ f \in L^1(\mathbb{R}^d) \cap L^2(\mathbb{R}^d) : \int_{\mathbb{R}^d} \|\xi\|_2^{2s} |\hat{f}(\xi)|^2 d\xi \leq (2\pi)^d R^2 \right\} \quad (1)$$

with smoothness parameter $s > 0$, radius $R > 0$, and where $\hat{f}$ denotes the Fourier transform of f , that is, $\hat{f}(\xi) := \int_{\mathbb{R}^d} f(x) e^{-ix^\top \xi} dx$ for all $\xi \in \mathbb{R}^d$. Our aim is to construct a test which

achieves the minimax rate of testing over $\mathcal{S}_d^s(R)$ (up to an iterated logarithmic term) and which does not depend on the smoothness parameter s of the Sobolev ball; such a test is called *minimax adaptive*.

As shown by Li and Yuan (2019, Theorems 3 and 5), the minimax rate of testing over the Sobolev ball $\mathcal{S}_d^s(R)$ is lower bounded as

$$\underline{\rho}(\mathcal{S}_d^s(R), \alpha, \beta, M) \geq C_0(M, d, s, R, \alpha, \beta) (m+n)^{-2s/(4s+d)} \quad (2)$$

for some constant $C_0 > 0$ depending on $\alpha, \beta \in (0, 1)$, $d \in \mathbb{N} \setminus \{0\}$ and $M, s, R \in (0, \infty)$. Their proof is an extension of the results of Ingster (1987, 1993b) and we provide more details in Appendix D. We later construct a test with non-asymptotic level α and show in Corollary 7 that its uniform separation rate over $\mathcal{S}_d^s(R)$ with respect to $m+n$ is at most $(m+n)^{-2s/(4s+d)}$, up to some multiplicative constant. This implies that the minimax rate of testing over the Sobolev ball $\mathcal{S}_d^s(R)$ with respect to $m+n$ is exactly of order $(m+n)^{-2s/(4s+d)}$.

2.2 Maximum Mean Discrepancy

As a measure between two probability distributions, we consider the kernel-based *Maximum Mean Discrepancy* (MMD—Gretton et al., 2007, 2012a). In detail, for a given Reproducing Kernel Hilbert Space $\mathcal{H}_k$ (Aronszajn, 1950) with kernel k , the MMD can be formalized as the integral probability metric (Müller, 1997)

$$\text{MMD}(p, q; \mathcal{H}_k) := \sup_{f \in \mathcal{H}_k : \|f\|_{\mathcal{H}_k} \leq 1} |\mathbb{E}_{X \sim p}[f(X)] - \mathbb{E}_{Y \sim q}[f(Y)]|.$$

Our particular interest is in a characteristic kernel k , which guarantees that we have $\text{MMD}(p, q; \mathcal{H}_k) = 0$ if and only if $p = q$. We refer to the works of Fukumizu et al. (2008) and Sriperumbudur et al. (2011) for details on characteristic kernels. It can easily be shown (Gretton et al., 2012a, Lemma 4) that the MMD is the $\mathcal{H}_k$ -norm of the difference between the mean embeddings $\mu_p(u) := \mathbb{E}_{X \sim p}[k(X, u)]$ and $\mu_q(u) := \mathbb{E}_{Y \sim q}[k(Y, u)]$ for $u \in \mathbb{R}^d$. Using this fact, a natural unbiased quadratic-time estimator for $\text{MMD}^2(p, q; \mathcal{H}_k)$ (Gretton et al., 2012a, Lemma 6) is

$$\begin{aligned} \widehat{\text{MMD}}_{\mathbf{a}}^2(\mathbb{X}_m, \mathbb{Y}_n; \mathcal{H}_k) &:= \frac{1}{m(m-1)} \sum_{1 \leq i \neq i' \leq m} k(X_i, X_{i'}) + \frac{1}{n(n-1)} \sum_{1 \leq j \neq j' \leq n} k(Y_j, Y_{j'}) \\ &\quad - \frac{2}{mn} \sum_{i=1}^m \sum_{j=1}^n k(X_i, Y_j). \end{aligned} \quad (3)$$

This is the minimum variance MMD estimator (Serfling, 1980, Section 5.1.4). As pointed out by Kim et al. (2022), this quadratic-time estimator can be written as a two-sample U -statistic (both of second order) (Hoeffding, 1992)

$$\widehat{\text{MMD}}_{\mathbf{a}}^2(\mathbb{X}_m, \mathbb{Y}_n; \mathcal{H}_k) = \frac{1}{m(m-1)n(n-1)} \sum_{1 \leq i \neq i' \leq m} \sum_{1 \leq j \neq j' \leq n} h_k(X_i, X_{i'}, Y_j, Y_{j'}) \quad (4)$$

where

$$h_k(x, x', y, y') := k(x, x') + k(y, y') - k(x, y') - k(x', y) \quad (5)$$

for $x, y, x', y' \in \mathbb{R}^d$. Writing the estimator $\widehat{\text{MMD}}_{\mathbf{a}}^2(\mathbb{X}_m, \mathbb{Y}_n; \mathcal{H}_k)$ as a two-sample U -statistic can be theoretically appealing but we stress the fact that it can be computed in quadratic time using Equation (3). The unnormalised version of the test statistic $\widehat{\text{MMD}}_{\mathbf{a}}^2(\mathbb{X}_m, \mathbb{Y}_n; \mathcal{H}_k)$ was also considered in the work of Fromont et al. (2012).

For the special case when $m = n$, Gretton et al. (2012a, Lemma 6) also propose to consider a different estimator for the Maximum Mean Discrepancy which is the one-sample second-order U -statistic

$$\widehat{\text{MMD}}_{\mathbf{b}}^2(\mathbb{X}_n, \mathbb{Y}_n; \mathcal{H}_k) := \frac{1}{n(n-1)} \sum_{1 \leq i \neq j \leq n} h_k(X_i, X_j, Y_i, Y_j). \quad (6)$$

Note that, unlike the estimator $\widehat{\text{MMD}}_{\mathbf{a}}^2(\mathbb{X}_n, \mathbb{Y}_n; \mathcal{H}_k)$, the estimator $\widehat{\text{MMD}}_{\mathbf{b}}^2(\mathbb{X}_n, \mathbb{Y}_n; \mathcal{H}_k)$ does not incorporate the terms $\{k(X_i, Y_i) : i = 1, \dots, n\}$. This means that the ordering of $\mathbb{X}_n = (X_i)_{1 \leq i \leq n}$ and $\mathbb{Y}_n = (Y_j)_{1 \leq j \leq n}$ changes the estimator $\widehat{\text{MMD}}_{\mathbf{b}}^2(\mathbb{X}_n, \mathbb{Y}_n; \mathcal{H}_k)$. So, when using this estimator, we have to assume we are given a specific ordering of the samples. While $\widehat{\text{MMD}}_{\mathbf{b}}^2$ has slightly higher variance than $\widehat{\text{MMD}}_{\mathbf{a}}^2$, computing $\widehat{\text{MMD}}_{\mathbf{b}}^2$ is computationally much faster than evaluating $\widehat{\text{MMD}}_{\mathbf{a}}^2$, as discussed in Appendix C.

The MMD depends on the choice of kernel, which we explore for our hypothesis tests.

3 Construction of tests and bounds

This section contains our main contributions. In Section 3.1, we introduce some notation along with technical assumptions for our analysis. We then present in Section 3.2 two data-dependent procedures to construct a single MMD test that makes use of some specific kernel bandwidth. Section 3.3 provides sufficient conditions under which this single MMD test is powerful when the difference in densities is measured in terms of the MMD and of the L^2 -norm. Based on these preliminary results, we prove an upper bound on its uniform separation rate over the Sobolev ball $\mathcal{S}_d^s(R)$ in Section 3.4, which shows that for a specific choice of bandwidth, the corresponding single MMD test is optimal in the minimax sense. However, the optimal single MMD test relies on the unknown smoothness parameter s of the Sobolev ball $\mathcal{S}_d^s(R)$, which motivates the introduction of our aggregated MMDAgg test in Section 3.5. Finally, we prove in Section 3.6 that MMDAgg is minimax adaptive over the Sobolev balls $\{\mathcal{S}_d^s(R) : s > 0, R > 0\}$.

3.1 Assumptions and notation

We assume that the sample sizes m and n are balanced up to a constant factor, meaning that there exists a positive constant $C > 0$ such that

$$m \leq n \quad \text{and} \quad n \leq Cm. \quad (7)$$

As can be seen in Equation (20), this assumption allows us to upper bound terms such as m^{-1} and n^{-1} by $(m+n)^{-1}$ up to a constant depending on C . The smaller this constant C